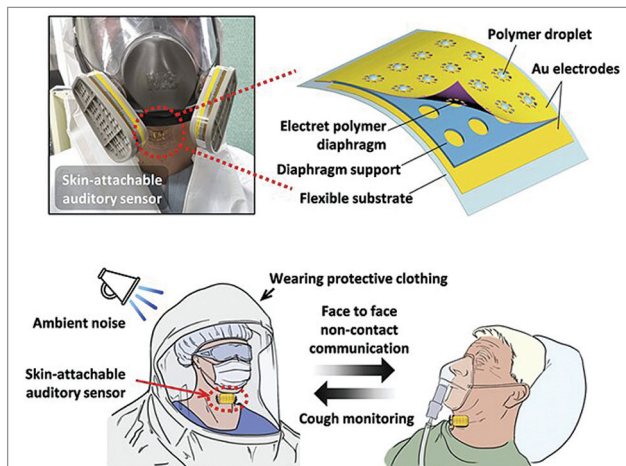




detailed and programmable spatio-temporal haptic feedback patterns, with 32 electrotactile simulation pixels on the side of the palm and a high spatial resolution of 0.279 pixels/cm<sup>2</sup> in the densest region. The advantage of this device is that it covers a wider surface of a user's hand (the whole hand) rather than focusing on one or more fingertips compared to electrotactile devices developed in the past.

### A skin-attachable auditory sensor that works well even in noisy conditions

Researchers at POSTECH have developed a skin-attachable auditory sensor that detects voices even in harsh, noisy environments. The electret-powered and hole-patterned polymer diaphragm is integrated with a skin-attachable auditory sensor, which reduces the necessity for a battery



A person with a polymer-based skin-attachable auditory sensor on the neck (upper left), a schematic diagram of the patch-type sensor (upper right), and an example of quarantine activities using such sensors (lower)  
(Source: <https://www.sciencedaily.com>)

to work. The sophisticated diaphragm structure based on MEMS technology gives the auditory sensor high wearability and portability. The diaphragm allows the sensor to detect the neck-skin vibration generated when a person speaks. Those using this sensor can hear sounds even in a concert hall or other noisy places, or when their faces are completely covered with a gas mask.

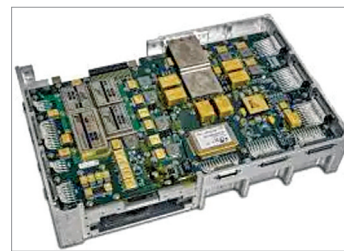
### A photonic chip that increases your data speed to petabits

A team of researchers has developed a photonic chip that is capable of handling nearly two petabytes of data per second. The chip developed by researchers with members from several institutions in Denmark, Sweden, and Japan is based on the use of photonics rather than electronics. The technology

implemented in their chip first splits an incoming data stream (from a laser) into 37 individual lines that travel across individual threads in a fibre cable. But before sending, the data in each of the 37 streams is split into 223 individual chunks, each corresponding to a unique part of the optical spectrum. This ensures the transfer of huge amounts of data.

### Hardware that helps measure 300,000 spectra/sec with superior quality

Decades ago, it took almost ten minutes to receive a single spectrum from a geological sample in the laboratory. To improve performance and speed, Alpha Data partnered with



The Focal Plane Interface Electronics - Digital (FPIE-D) supplied by Alpha Data  
(Source: <https://businessfortnight.com>)

NASA's Jet Propulsion Laboratory (JPL) and developed high-performance FPIE-D hardware that supports EMIT's imaging spectrometer to measure 300,000 spectra per second with superior quality. Alpha Data's development of EMIT's FPIE-D board,

in conjunction with the Jet Propulsion Laboratory (JPL), required extensive original research and development, which the company says will be transferable to future projects requiring space-proven data acquisition, cloud screening, compression, storage, and downlink systems. EMIT was installed on the International Space Station's truss.

### A lidar sensor that extends the viewing angle to 360 degrees

A team of researchers from POSTECH has developed a lidar sensor made from metasurface that extends the viewing angle to 360°, enabling autonomous vehicles to detect unpredictable conditions on the road. This ultra-small lidar sensor is built from the metasurface, an ultra-thin flat optical device that is only one-thousandth the thickness of a human hair strand. Utilising the metasurface can greatly expand the viewing angle of the lidar to recognise objects three-dimensionally. The research team extended the lidar sensor's viewing angle to 360° by modifying the design and periodically arranging the nanostructures that make up the metasurface. It is possible to extract three-dimensional information about objects in 360° regions by scattering more than 10,000 dot arrays (light) from the metasurface onto the objects and photographing the irradiated point pattern with a camera.